

Chitwan Saharia

Research Engineer, Google Brain Toronto

✉ sahariac@google.com

Education

B.Tech. Compute Science and Engineering

2015-19

Indian Institute of Technology, Bombay, India

Professional Experience

Google Brain

Toronto

Research Engineer / AI Resident

Oct 2019 - Current

- Working on generative modeling of discrete and continuous structured data.
- Research Areas: Image and text generation, Non-autoregressive models, Diffusion models.

Mila

Montreal

Research Intern

May-August 2018/2019

- Worked on benchmarking learning algorithms for Grounded Language Learning.
- Research Areas: Imitation Learning, Goal Conditioned RL.

Publications

- **Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding**
Chitwan Saharia*, William Chan*, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S. Sara Mahdavi, Rapha Gontijo Lopes, Tim Salimans, Jonathan Ho, David J Fleet, Mohammad Norouzi*
Pre-print arXiv, 2022
- **Deblurring via Stochastic Refinement**
Jay Whang, Mauricio Delbracio, Hossein Talebi, **Chitwan Saharia**, Alexandros G. Dimakis, Peyman Milanfar
Computer Vision and Pattern Recognition (CVPR), 2022 [ORAL]
- **Palette: Image-to-Image Translation Models**
Chitwan Saharia, William Chan, Huiwen Chang, Chris A. Lee, Jonathan Ho, Tim Salimans, David J. Fleet*, Mohammad Norouzi*
SIGGRAPH Conference (2022)
Workshop on Deep Generative Models and Applications, NeurIPS, 2021
- **Cascaded Diffusion Models for High Fidelity Image Generation**
Jonathan Ho*, **Chitwan Saharia***, William Chan, David J. Fleet, Mohammad Norouzi, Tim Salimans
Journal of Machine Learning Research (JMLR), 2022
- **Image Super-Resolution via Iterative Refinement**
Chitwan Saharia, Jonathan Ho, William Chan, David J. Fleet, Mohammad Norouzi
Pattern Analysis and Machine Intelligence (PAMI), 2022

- **Non-Autoregressive Machine Translation with Latent Alignments**
Chitwan Saharia*, William Chan*, Saurabh Saxena, Mohammad Norouzi
Empirical Methods in Natural Language Processing (EMNLP), 2020
- **Imputer: Sequence Modelling via Imputation and Dynamic Programming**
William Chan, Chitwan Saharia, Geoffrey Hinton, Mohammad Norouzi, Navdeep Jaitly
International Conference on Machine Learning (ICML), 2020
- **Combating False Negatives in Adversarial Imitation Learning for Instruction Following**
K. Zolna*, Chitwan Saharia*, L. Boussieux*, David H, Maxime CB, D Bahdanau, Yoshua Bengio
AAAI Conference (Student Abstract Track), 2020
NeuRIPS Deep Reinforcement Learning Workshop, 2020
- **A Tale of Two Modalities for Video Captioning**
P. Joshi*, Chitwan Saharia*, Vishwajeet Singh, D. Gautam, G. Ramakrishnan, P. Jyothi
Workshop on Multi-modal Video Analysis & Moments in Time Challenge, ICCV, 2019
- **BabyAI: First Steps Towards Grounded Language Learning With a Human In the Loop**
Maxime CB, Dzmitry Bahdanau, Salem L, Lucas W, Chitwan Saharia, Thien HN, Yoshua Bengio
International Conference on Learning Representations (ICLR) 2019

Other Research Experiences

Bachelor's Thesis [Talk]

IIT Bombay

Advisors : Prof. Preethi Jyothi and Prof. Ganesh Ramakrishnan

August 2018 - May 2019

- Analyzed the role of audio and visual modalities in Video Captioning.

R&D Project [Report]

IIT Bombay

Advisor : Prof. Suyash Awate

August 2018 - December 2018

- Analyzed impact of calibration techniques on cell segmentation tasks.

Seminar [Report]

IIT Bombay

Advisor : Prof. Pushpak Bhattacharyya

January 2018 - April 2018

- Explored the literature on machine learning and rule based models for sentiment analysis.

Scholastic Achievements

- Secured All India Rank 148 in IIT JEE Examination (2015) among 1.5 million thousand candidates.
- Recipient of Kishore Vaigyanik Protsahan Yojna Fellowship instituted by the Department of Science and Technology, Govt. of India.

Positions of Responsibility

Teaching Assistant

- Object Oriented Programming at IIT BombayX
- Computer Programming at IIT Bombay

Reviewer

- International Conference on Learning Representations (ICLR) [2021, 2022]
- Neural Information Processing Systems (NeurIPS) [2021]